



OPEN

DATA DESCRIPTOR

A Figure-Text Pair Dataset from Map-related Literature for Multimodal Model Research

Ming Liang¹, Hui Li¹, Yunqiang Zhu^{2,3}✉ & Jinqiu Zhang¹✉

High-quality multimodal datasets are essential for developing vision-language models, yet publicly available figure-text resources in specialized scientific domains remain limited. To address this gap, we present a large-scale figure-text pair dataset constructed from map-related scientific literature (FTPD-ML). The dataset was derived from 96,859 scientific publications in cartography, geography, remote sensing, and related disciplines, and provides 75,702 publicly redistributable figure-text pairs under Creative Commons Attribution (CC BY) licenses with multiple levels of textual representations, including original figure captions, standardized caption variants, and contextual paragraph descriptions, together with bibliographic metadata. To ensure compliance with copyright and licensing requirements, records from non-redistributable publications are represented by metadata only. The dataset supports a range of multimodal research tasks, including image-text retrieval, image captioning, scientific document understanding, and cartography-oriented vision-language studies.

Background & Summary

The rapid iterative evolution of multimodal large models has achieved a leap from theoretical exploration to widespread practical applications^{1–5}, while the scarcity of high-quality multimodal data severely hinders the models' ability to profoundly comprehend cross-modal semantics^{6,7}. As a result, how to construct high-quality multimodal datasets has become a critical issue for large model research^{8–10}.

Scientific and technical literature as a valuable source of knowledge contains a vast number of high-quality textual descriptions, figures, and rigorous interpretations of diverse scientific discoveries^{11,12}. Within these articles, figure-text pairs—comprising illustrations and their accompanying textual descriptions—not only provide visual representations of scientific research outcomes, but also play an irreplaceable role in explaining technical principles and conveying complex knowledge. These figure-text data incorporate cross-modal structures that bridge visual and linguistic information, making them valuable resources for multimodal representation learning^{13–17}. As such, they represent an essential and high-quality corpus for training multimodal large language models. However, due to formatting constraints, typography, and layout in scientific papers, figures and their corresponding texts often lack a strong one-to-one correspondence, making them unsuitable for direct use in large model training. Consequently, constructing large-scale, high-quality figure-text datasets from scientific literature remains a significant challenge because figures and their associated textual descriptions are often difficult to accurately identify and associate within complex document layouts^{18–20}.

Currently, transferring knowledge across disciplines remains a challenge for the creation of multimodal models^{21,22}, as figure-text data from different topic fields frequently have their own unique expressive paradigms and semantic systems^{23,24}. This significantly raises the requirements for models' domain adaptation capabilities. For the subject of cartography, the map image frequently incorporates several semantic aspects, including geocoding, legend symbols, and spatial linkages. Constructing a figure-text pair dataset embedded with cartographic domain knowledge is essential for enabling deep comprehension of such complex information. Therefore, datasets that explicitly preserve cartographic semantic information, including thematic content, spatial relationships, and legend symbols, are important for supporting multimodal research in this domain^{25,26}.

¹School of Computer Science, South China Normal University, Guangzhou, 510631, China. ²College of Aerospace Intelligence, Hainan University, Haikou, 570228, China. ³State Key Laboratory of Resources and Environment Information System, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing, 100010, China. ✉e-mail: 997455@hainanu.edu.cn; zjq@scnu.edu.cn

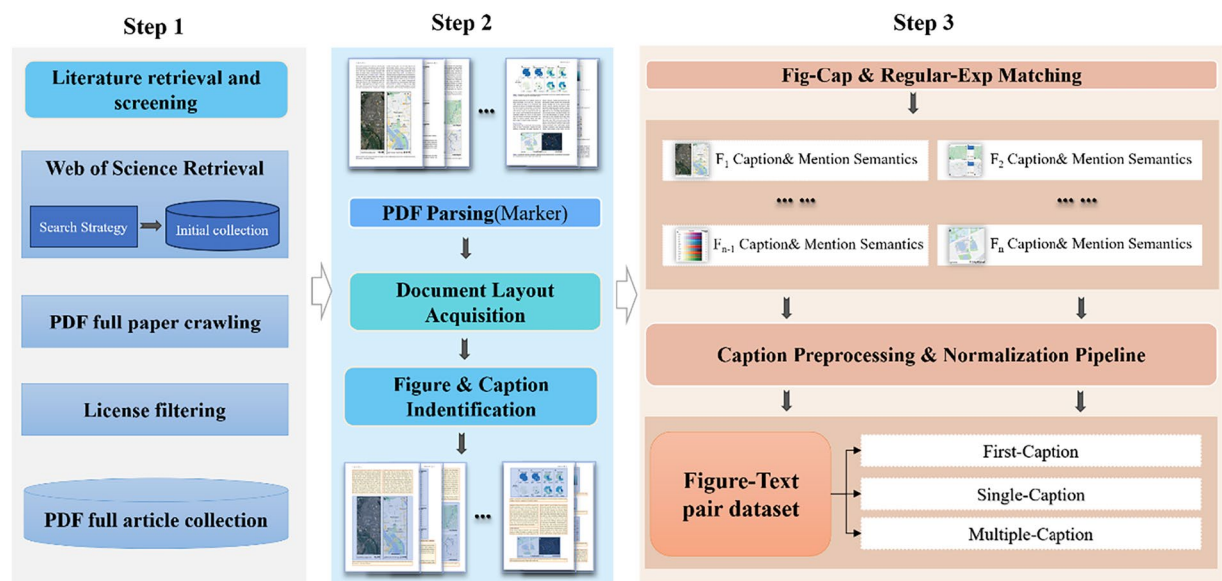


Fig. 1 Workflow of figure-text pair dataset construction.

To facilitate multimodal research on scientific figures in cartography, we constructed the Figure-Text Pair Dataset from Map-related Literature (FTPD-ML). The dataset was constructed by screening 96,859 map-related scientific publications and extracting approximately 500,000 figure-text pairs. Each record includes multi-granularity textual descriptions and article metadata to facilitate downstream multimodal applications.

Representative downstream tasks, including image-text retrieval and image captioning, were conducted to demonstrate the usability of the FTPD-ML dataset. These validation experiments illustrate that FTPD-ML provides effective supervision for representative multimodal tasks and can serve as a reusable benchmark resource for future cartography-related vision-language research.

The remaining sections of this article are organized as follows. Section 2 describes the details of the construction method. Section 3 contains dataset records and section 4 contains the data overview. Section 5 presents detailed technical validation that confirms the validity of the dataset. Finally, Section 6 provides usage notes.

Methods

Figure 1 illustrates the overall framework and workflow for constructing this dataset. The objective is to curate a collection of figure-text pairs from map-related literature, tailored specifically for developing multimodal large models in cartography. The construction pipeline begins with PDF documents collected from map-related scientific literature. PDF documents are first parsed into Markdown format²⁷, with content saved separately as image and text files. Figures, captions, and contextual paragraphs are subsequently extracted and associated to construct initial figure-text pairs. Based on these initial pairs, and in accordance with the requirements of multimodal large models, standardized preprocessing procedures are applied to produce the released dataset.

Literature Retrieval and Domain-Relevance Screening. To improve the domain relevance of the source literature corpus and ensure the reproducibility of the retrieval process, this study first constructed a composite search strategy targeting cartography and spatial information science in the Web of Science database. The search query was composed of core cartographic concepts, geospatial information concepts, and spatial application scenario terms, with parentheses explicitly defining the logical relationships among different conceptual groups.

Given that certain retrieval terms are widely used across multiple disciplines and may therefore extend beyond the scope of geospatial research in their semantic coverage, a subsequent domain-relevance filtering procedure was designed after the initial retrieval to enhance the specificity and accuracy of the dataset sources.

The filtering process adopts a hierarchical refinement strategy. Domain relevance is assessed based on journal source, article title, abstract, and keywords to identify studies with a clear background in cartography or spatial information science. For records with ambiguous disciplinary attribution, a further comprehensive judgment is conducted by integrating abstract-level context and spatial semantic features. This procedure effectively reduces false positives introduced by keyword-based retrieval while preserving closely related interdisciplinary studies, including remote sensing, geographic information systems (GIS), geovisualization, land use/land cover mapping, and spatial analysis.

The overall framework of the retrieval and filtering pipeline is illustrated in Fig. 2. Detailed statistics of the filtering results and the manual validation process are provided in the Data Records and Technical Validation sections, respectively.

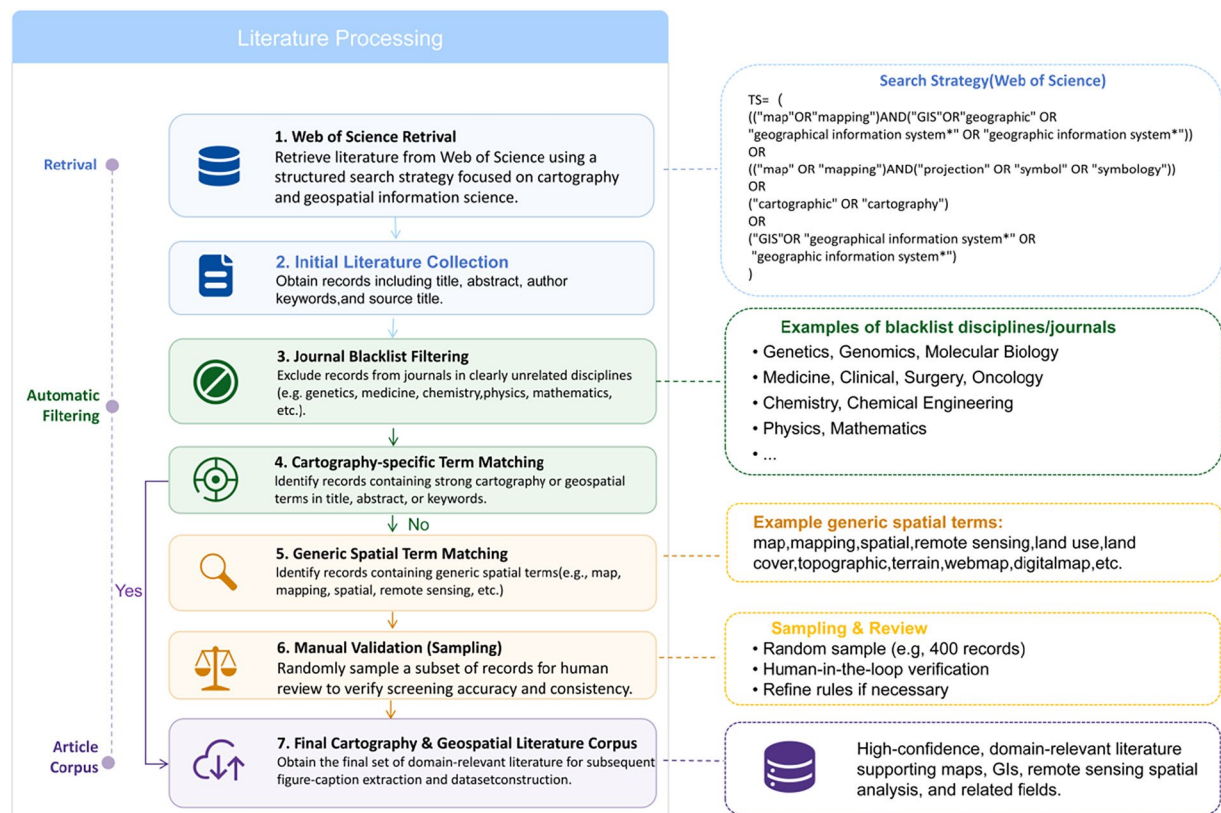


Fig. 2 Literature retrieval and domain relevance screening workflow.

License Identification and Public Release Strategy. FTPD-ML is derived from figures, figure captions, and surrounding contextual text extracted from published scientific literature. To ensure that all publicly distributed records comply with copyright and redistribution requirements, the license associated with each publication was retrieved through the Unpaywall service²⁸. Only publications explicitly licensed under CC-BY were considered eligible for public redistribution. Publications with restrictive licenses (e.g., CC BY-NC, CC BY-ND), unknown licenses, or without redistributable licenses were excluded from the released figure-text dataset.

The CC-BY status was determined based on licensing information retrieved from Unpaywall. For articles with missing reliable metadata or where the CC-BY license could not be unambiguously verified, the corresponding figures and textual content were excluded from the publicly redistributable portion of the dataset.

Based on the identification results, a tiered data-sharing strategy was adopted. For samples originating from CC-BY licensed publications and passing the domain-relevance filtering process, figures, captions, contextual text, and associated metadata are made available, subject to the terms of the original license and attribution requirements. For samples derived from non-redistributable publications, the dataset does not release original images or extended textual content; instead, only essential metadata, source identifiers, and derived statistical features are provided. The detailed definition of the sharing scope and released components is further described in the Data Records section.

PDF File Parsing and Figure-text Pair Extraction. PDF (Portable Document Format) is a cross-platform file format created by Adobe Systems that has evolved into the international standard for storing and distributing academic papers, technical and official documents²⁹. Its primary advantage lies in its ability to preserve the original layout of documents—including text, figures, tables, mathematical formulas, and other elements—ensuring consistent presentation across different devices and operating systems. However, this fixed formatting poses challenges for structured data extraction required in large model training, and various PDF-oriented parsing tools have been developed, such as Llaparse³⁰ and Mathpix³¹. PDF documents were converted into Markdown format using Marker³². This format preserves the document's chapter structure and metadata, which facilitates subsequent analysis of graphical structure and content.

Extraction of Figure and Caption. To achieve structured integration and semantic pairing of visual and textual information, we started by analyzing the typographic features of PDF documents after conversion to Markdown format. It is observed that, in Markdown documents, figures are typically embedded using the syntax of “”, and captions immediately follow in the form of “Figure X...”. Based on these structural characteristics, figures and their corresponding captions were automatically associated using rule-based matching procedures.

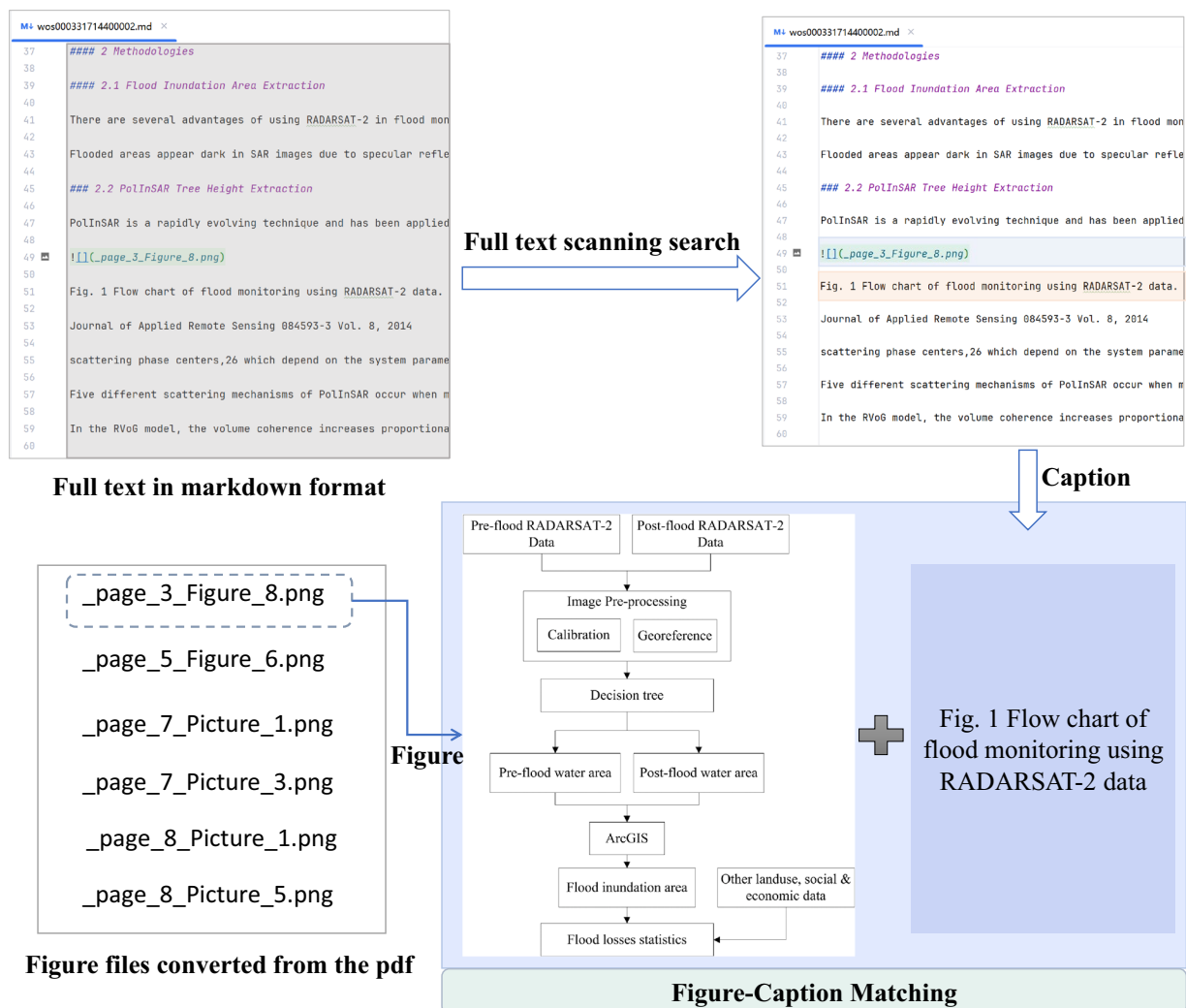


Fig. 3 Schematic diagram of figure-text pair extraction. Reproduced from Study of RADARSAT-2 synthetic aperture radar data for observing sensitive factors of global environmental change, by Guo, H., Liu, G., Liao, J., Li, X., Zhang, L., Shen, G., Fu, W., & Sun, Z., Journal of Applied Remote Sensing, 8, 084593 (2014) (<https://doi.org/10.1117/1.JRS.8.084593>). CC BY 4.0⁴⁰.

The primary matching criterion is as follows: if a paragraph of text directly follows a figure tag and begins with “Figure” or “Fig.”, it is assumed that the figure and the paragraph constitute a valid figure-text pair. Additionally, paragraphs that do not contain figure markers or do not conform to the expected Figure-Caption format are excluded, which significantly improves pairing accuracy. Figure 3 presents the schematic workflow of the figure-text extraction process.

The rule-based procedure enables efficient large-scale figure–caption association while maintaining one-to-one correspondence between figures and captions. The resulting correspondence between figures and captions provides essential data support for subsequent multimodal semantic modeling.

Extraction of Contextual Paragraph Text. Contextual paragraphs are linked based on the established figure-caption pairs. Figure number information is first parsed from captions, and corresponding figure citations are located within the main text of each article. Sentences or paragraphs containing these citations are then extracted as contextual descriptions associated with the target figures.

The extraction procedure is designed to be compatible with common forms of figure referencing and is standardized to account for variations in language (English and Chinese), capitalization, and abbreviation formats.

For cases where multiple contextual paragraphs are associated with the same figure, all relevant passages are retained in order to preserve, as comprehensively as possible, the explanatory content, methodological descriptions, and result discussions related to the figure in the source literature. The accuracy of the contextual linking process is further evaluated through manual sampling validation, with detailed results reported in the Technical Validation section.

Text Cleaning and Normalization. To enhance the quality and consistency of text descriptions, a unified cleaning and normalization procedure was applied across all constructed datasets. Since figure captions are automatically extracted, they often exhibit inconsistencies in formatting, redundant symbols, embedded formulas, and similar issues, which may introduce semantic noise or inefficient token utilization if directly fed into downstream models. Therefore, text cleaning and normalization are essential for building a high-quality multimodal corpus. In this study, the cleaning process is implemented through regular expressions and consists of three main steps:

Removal of Non-Informative Content: Automatic identification and elimination of literature citations (e.g., [1], [12]), LaTeX-style formulas (e.g., $x = y^2$), and redundant special characters (e.g., non-linguistic symbols and meaningless marks). This step removes noise that does not contribute to model learning.

Character and Punctuation Filtering: Retention of Chinese and English characters, Arabic numerals, and common punctuation in both languages (e.g., periods, commas, semicolons, exclamation marks). Symbols that carry no semantic value—such as repeated dashes, uncommon Unicode characters, and other non-informative symbols—are filtered out. This ensures clear text structure and effective tokenization.

Whitespace and Format Standardization: Unification of spaces between Chinese and English text, removal of extra spaces, and normalization of punctuation spacing. This step enforces a consistent input structure, reduces token-segmentation ambiguity, and improves token-usage efficiency.

Through this process, the textual component corresponding to each figure is significantly improved in semantic accuracy, structural integrity, and model usability, establishing a more stable and reliable textual foundation for subsequent model training and evaluation.

Figure-text Pair Classification and Preprocessing. Figure captions across different types of literature vary widely in terms of text length, expression style, and semantic structure. Some captions are concise descriptions, while others contain complex phrasing or nested, multi-layered information, including background context, methodological explanations, and result summaries. This semantic heterogeneity poses a challenge for multimodal modeling: directly using raw figure captions may lead to inconsistencies in semantic granularity and text length among training samples, thereby undermining the model's generalization ability and training stability.

To enhance the semantic standardization and structural consistency of figure captions, we process the original captions (Figure-Captions) based on text length and syntactic structure into three categories drawing on the processing strategies proposed by Hsu³¹. The first category, referred to as First-Caption, extracts only the first sentence of the original figure caption as the standardized output. This approach is well-suited for general scenarios involving a large number of figures with lengthy captions and complex semantic structures, as it effectively compresses the text length while preserving the core semantic content. The second type of figure-text pair, referred to as Single-Caption, includes only those figures whose original figure captions consist of only one single sentence. This category is particularly suitable for figures in academic papers that feature concise, focused descriptions and a compact presentation of information. The third category, referred to as Multiple-Caption, applies to figures accompanied by multi-sentence captions. This type is suitable for high-information-density figures that require maintaining contextual coherence and preserving detailed semantic content.

With the above classification strategy, tailored modeling can be performed based on the complexity of the figure content, thereby reducing the impact of redundant noise while ensuring sufficient semantic coverage. Meanwhile, this standardized approach enhances the controllability and consistency of the dataset structure, facilitating more efficient use and evaluation in downstream tasks such as image matching and image captioning.

The multi-granularity semantic labeling technique, which includes contextual paragraph text, First-Caption, Single-Caption, and Multiple-Caption, provides multiple textual representations that support diverse downstream multimodal applications. Table 1 presents an example from the figure-text dataset after standardization. The first line is a figure extracted from map-related literature and Figure-Caption stands for its original caption. The remaining entries illustrate different caption representations associated with the figure. It can be seen that the Multiple-Caption is blank because the original figure caption has only one sentence.

Data records

The FTPD-ML dataset is publicly available through the Zenodo repository³³ and is organized into multiple components to facilitate data access, reuse, and downstream multimodal research. The overall data repository is organized in a hierarchical structure, as illustrated in Fig. 4.

The publicly released component contains figure-text pairs originating from publications licensed under CC-BY and passing the map-related relevance filtering procedure. In addition, metadata covering the full dataset, and source indexing information are provided. For records originating from publications without eligible redistribution licenses, only metadata permitted under compliance constraints are released. Original images and extended body text are not included, in order to ensure that dataset distribution adheres to copyright and licensing requirements. Figure-text samples, article-level metadata, and example datasets are stored in a decoupled manner to achieve separation between data content and metadata.

Figure-Text Records. The Figure_Text_Pairs component contains the primary multimodal data used for downstream applications. Each record corresponds to one extracted scientific figure together with its associated textual descriptions. Besides the original figure caption and contextual paragraph extracted from the source publication, the dataset provides three standardized caption variants with different semantic granularities (First-Caption, Single-Caption, and Multiple-Caption), allowing users to select appropriate textual representations for different multimodal applications.

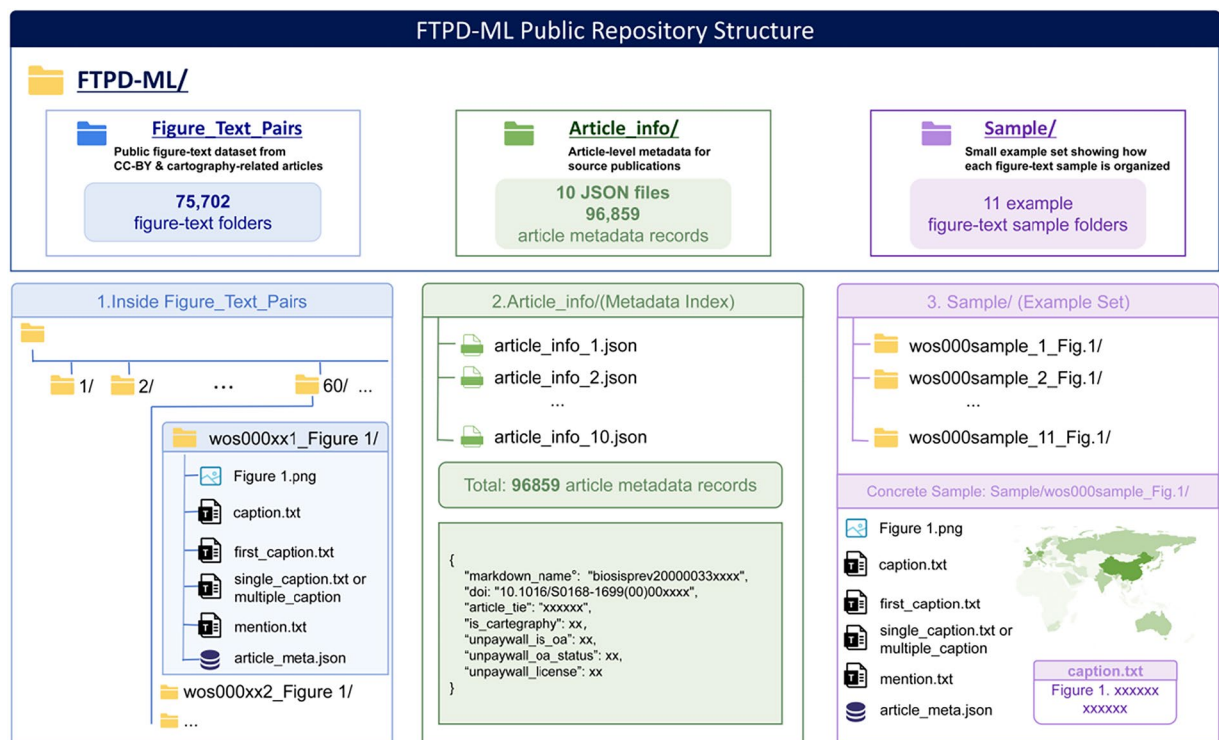


Fig. 4 FTPD-ML Public Repository Structure.

Each figure–text record is accompanied by an `article_meta.json` file that stores metadata of its source publication, including the article title, DOI (where available), license information, Open Access status, domain relevance information, and additional classification attributes. Specifically, the `class_name` field records the image category assigned to the figure, while the `is_cartography` field indicates whether the source article is classified as a map-related publication. These metadata establish explicit links between individual figure–text records and their original publications, while providing information on image types and domain classification. Together, they support provenance tracking, quality assessment, and reproducible reuse of the released dataset.

Article-Level Metadata. The `Article_Info` component stores publication-level metadata for every processed document. These records include identifiers such as article-title, DOI, Open Access status and license. Article-level metadata enable users to associate individual figure–text pairs with their original scientific publications and support bibliometric or document-level analyses.

For publications that cannot be redistributed because of copyright restrictions, only article metadata are retained within the released repository, while figure images and textual contents remain unavailable.

Sample Records. The `Sample` component provides a selection of representative examples to illustrate the data fields and hierarchical organization of the broader Figure–Text Pairs structure. This subset is specifically designed to offer readers a lightweight preview, allowing them to inspect the file formats and formatting styles without the need to download the entire dataset.

Released Dataset. The FTPD-ML repository provides publicly accessible figure-text records originating from CC-BY licensed map-related publications together with their associated metadata. Each released record contains figure images, textual descriptions and bibliographic information. Records derived from publications without eligible redistribution licenses are represented by metadata only. The overall composition and statistical characteristics of both the complete corpus and the released subset are presented in the following Data Overview section.

Data Overview

This section summarizes the statistical characteristics of both the complete corpus constructed during dataset development and the publicly released subset. Although only Open-license (CC-BY) figure–text pairs are distributed through the FTPD-ML repository, the complete construction pipeline processed 96,859 scientific publications and extracted 497,951 figure–text pairs. Among these, 75,702 figure-text pairs originate from CC-BY licensed publications and satisfy the map-related relevance criteria. The following sections describe the composition and distribution of these data from multiple perspectives.

Extracted Figure-Text Pairs from Map Related Literature. The complete FTPD-ML corpus contains multiple textual representations for each extracted figure-text record, enabling downstream tasks with different

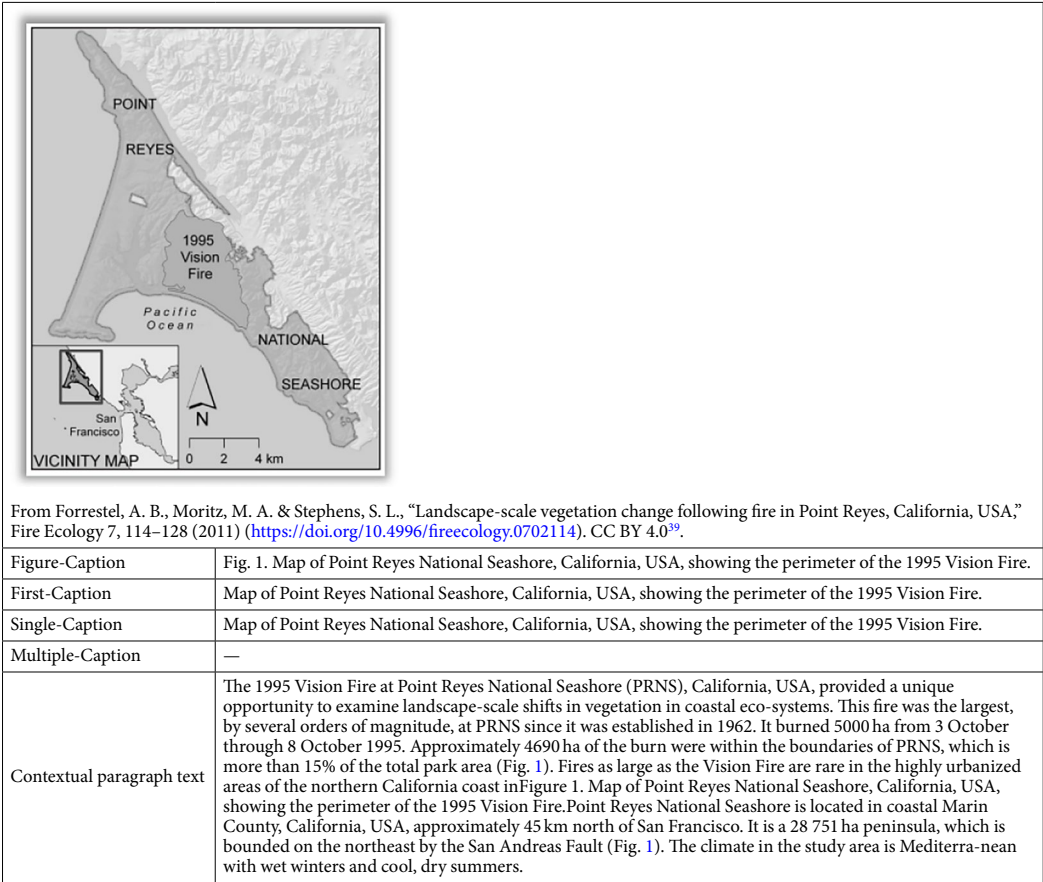


Table 1. Example of Figure-Text pair data.

Type	Total	Avg. Caption Length
Contextual paragraph text	497951	199
Figure-Caption	497951	32
First-Caption	497951	16
Single-Caption	240005	17
Multiple-Caption	256328	46

Table 2. Figure count and average text length in FTPD-ML.

levels of semantic granularity. As shown in Table 1, each figure has five different texts matching it, including contextual paragraph text, original Figure-Caption, and three subsets categorized by the Figure-Caption. Table 2 presents the records number and average text length in FTPD-ML. Both the Figure-Caption and the contextual paragraph text consist of 497,951 records, maintaining a strict one-to-one correspondence with figures. Based on variations in length and structure, the Figure-Caption is further classified into three different types of caption patterns: First-Caption (497,951 items), Single-Caption (240,005 items), and Multiple-Caption (256,328 items).

Open-license Subset and Release Scope. To quantify the impact of copyright constraints on public data release, we further summarize the distributions of Open-license and domain relevance at both the publication and figure-text levels, as summarized in Table 3.

At the publication level, a total of 96,859 source articles were collected. Among them, 33,857 articles (35.0%) were identified as Open Access, while 19,369 articles (20.0%) were classified as CC-BY licensed. After domain-relevance filtering, 55,057 articles (56.8%) were determined to be relevant to cartography or spatial information science, whereas 41,802 articles (43.2%) were excluded from the domain-relevant subset.

At the figure-text sample level, a total of 497,951 samples were extracted from all source articles. Among these, 298,053 samples (59.9%) originated from map-related publications. Furthermore, 75,702 samples (15.2% of the total) simultaneously satisfy both CC-BY licensed and domain-relevance criteria. These samples constitute the publicly downloadable figure-text subset of FTPD-ML. For samples derived from non-redistributable

Unit	Category	Count	Proportion
Articles	ALL	96859	100%
Articles	Open Access	33857	35.0%
Articles	License(cc-by)	19369	20.0%
Articles	map-related	55057	56.8%
Articles	Non-cartography-related	41802	43.2%
Figure-text pairs	ALL	497951	100%
Figure-text pairs	map-related	298053	59.9%
Figure-text pairs	License(cc-by) and map-related	75702	15.2%

Table 3. Statistics of Open-license Status for Source Articles and Figure-Text Samples.

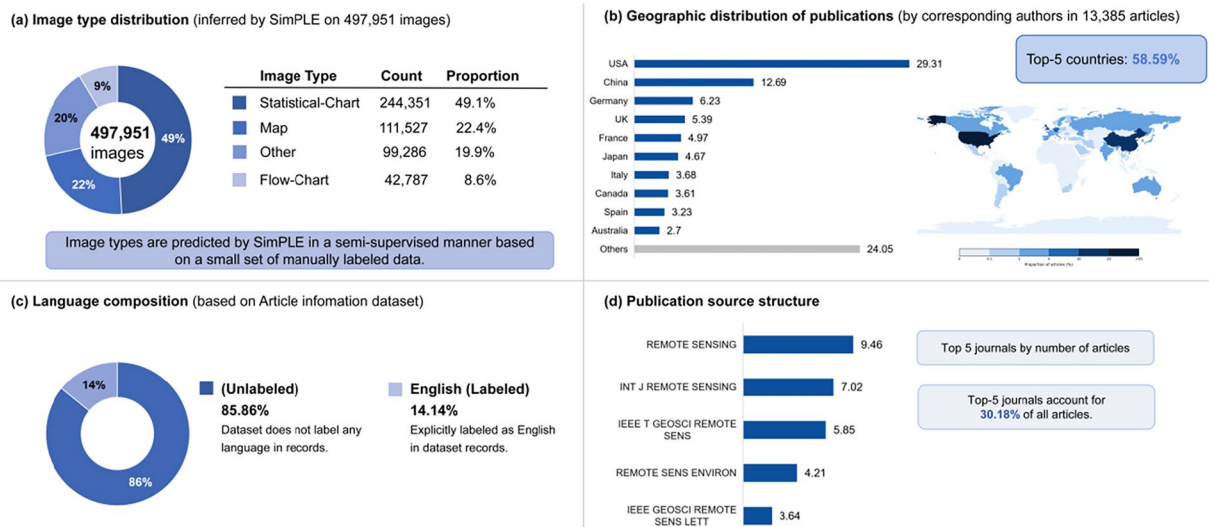


Fig. 5 Dataset composition overview of FTPD-ML. **(a)** Distribution of image types predicted by SimPLE on 497,951 images. **(b)** Geographic distribution of publications by corresponding author in 13,385 articles. **(c)** Language composition based on Web of Science records. **(d)** Publication source structure, including top journals(annotated records only).

publications, original images, full captions, and extended contextual text are not released; only article-level metadata are made publicly available.

At the source article level, each publication is annotated according to its Open-license status. For articles identified as CC-BY licensed, those that also pass the cartography and spatial information science relevance screening are included in the publicly released subset, together with their figures, captions, contextual text, and associated metadata. For non-redistributable articles, original figures, full captions, and extended contextual text are withheld due to copyright constraints, and only metadata are retained.

Dataset Composition and Structural Characteristics. To characterize the overall composition of the FTPD-ML dataset, this study conducts a statistical analysis based on 96,859 source articles from four perspectives: figure type statistics, geographic distribution, linguistic composition, and publication source statistics. The objective of this analysis is to describe the structural distribution patterns of the dataset within the field of cartography and spatial information science, thereby providing contextual evidence for downstream model training and cross-task applications.

Figure type statistics. Using the SimPLE semi-supervised learning framework, 497,951 figures were classified into four categories: map, statistical chart, flow chart, and other. The results indicate pronounced imbalances across categories, with statistical charts representing the dominant class, while flow charts account for a relatively smaller proportion. This distribution reflects the natural frequency of image types in scientific literature rather than any form of balanced or controlled sampling. The overall distribution of image types is illustrated in Fig. 5(a).

Geographic Distribution. From the perspective of country/region of publication, the dataset exhibits a typical global research distribution pattern, primarily concentrated in countries with strong scientific output. The United States, China, Germany, the United Kingdom, and Italy account for the largest proportions. The top-10 countries collectively contribute approximately 75.95% of the dataset, while the top-5 countries account for

approximately 58.59%. This distribution reflects the empirical structure of global geospatial information science research and is also influenced by the indexing coverage of the Web of Science database. The country/region distribution is shown in Fig. 5(b).

Linguistic Composition. The linguistic analysis shows that the dataset is overwhelmingly composed of English-language publications. In our article information database, most English-language articles do not explicitly include language metadata; consequently, approximately 85.86% of records have missing language fields, while the remaining entries are explicitly labeled as English. The distribution of language fields is illustrated in Fig. 5(c).

Publication Source Statistics. The dataset is primarily derived from leading journals in remote sensing, geographic information science, and Earth observation, demonstrating a strong disciplinary concentration. Representative journals include Remote Sensing, International Journal of Remote Sensing, and IEEE Transactions on Geoscience and Remote Sensing. The top five journals account for approximately 30.18% of the dataset. The distribution of journals is shown in Fig. 5(d).

Overall, these distribution characteristics are not introduced by sampling bias, but rather emerge from the natural distribution of scientific literature and the indexing mechanisms of the Web of Science database. Consequently, while maintaining large-scale diversity, the dataset exhibits clear domain consistency in its statistical structure, providing a representative real-world foundation for downstream multimodal learning tasks.

Technical Validation

The technical validation of FTPD-ML is conducted from four perspectives: domain relevance, figure-caption extraction accuracy, figure-text semantic alignment, and image quality assessment. The objective is to evaluate the reliability, consistency, and reusability of the dataset. In addition, figure-text matching and image captioning experiments are included as application-level validations to assess the dataset's capability to support downstream multimodal tasks.

Domain Relevance Validation. To assess the accuracy of the domain filtering process, 1,000 articles were randomly sampled from the automatically filtered corpus for manual verification. Two annotators with backgrounds in cartography or geographic information science independently evaluated each sample based on title, abstract, keywords, and journal source, and categorized them into three classes: highly relevant, moderately relevant, and weakly relevant or irrelevant. In cases of disagreement, a third annotator was involved for adjudication.

The results indicate that the overall domain relevance rate of the filtered corpus is 85.2%, with 62.8% of articles classified as highly relevant, 22.4% as moderately relevant, and 14.8% as weakly relevant or irrelevant. The inter-annotator agreement, measured using Cohen's kappa, reaches 0.84, indicating substantial consistency in domain relevance annotation.

Figure-Caption Extraction Validation. To evaluate the accuracy of the figure and caption extraction pipeline, a manually annotated validation set was constructed. A stratified random sample of 1,000 figure-caption pairs was selected across publication years, journal sources, and document layout types, covering single figures, multi-panel figures, long captions, non-standard captions, cross-page captions, and complex layouts.

Two annotators independently assessed figure extraction completeness, caption matching correctness, and caption text integrity. Disagreements were resolved by a third annotator. The results show that figure-text pair extraction coverage reaches 91.7%, and caption text completeness is 92.4%. The main sources of error include multi-panel figure decomposition, cross-page caption association, incomplete figure boundary detection in scanned PDFs, and a small number of non-standard caption formats.

Figure-text alignment. Figure-text alignment is a critical stage in the construction of a multimodal dataset, as its accuracy directly influences the effectiveness of semantic learning in subsequent model training. Since most of the original data in the literature originates from figures and captions, there is a certain proportion of mismatches, extraction errors, or semantic inconsistencies. Therefore, it is essential to conduct a statistical evaluation of the figure-text alignment to ensure the reliability of the pairings.

In this study, we employ the CLIP (Contrastive Language-Image Pretraining)³⁴ model proposed by OpenAI to evaluate the semantic alignment of figure-text pairings, leveraging its strong performance in zero-shot semantic matching tasks. Specifically, we use the CLIP-ViT/B-32 model to extract semantic features from both the figures and their corresponding annotated texts. The cosine similarity between these feature vectors is then calculated to produce a figure-text alignment score, which quantifies their closeness within a shared semantic space. Higher scores indicate stronger semantic associations between the figure and text.

Using a corpus of 100,000 figure-caption pairs, we analyzed the similarity score distribution and, following manual review, establishes a filtering threshold of 0.2 for low similarity (Fig. 6). This threshold is supported by three key findings: 1) the CLIP model shows that most valid pairs score between 0.2 and 0.4; 2) manual inspection of about 1,000 pairs reveals that scores below 0.2 often indicate semantic mismatch, extraction errors, or caption corruption; and 3) the 0.2 threshold provides a stringent yet practical criterion that balances data quality with acceptable misidentification rates.

When this threshold is applied to the dataset reveals that approximately 0.36% of samples fall below 0.2. Manual review of these low-scoring samples shows that only about 0.02% constitute clear errors—such as completely unrelated figures and captions or missing annotations. The majority are actually reasonable semantic

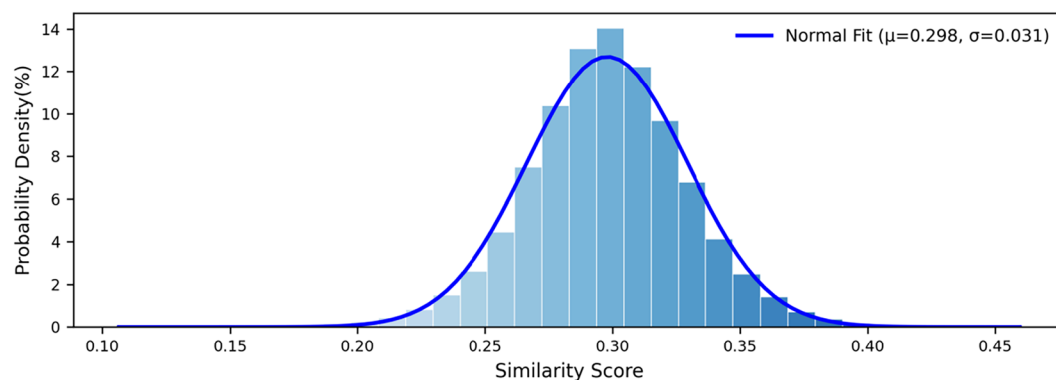


Fig. 6 Distribution of figure-text similarity scores by CLIP.

matches underestimated by the model, primarily due to overly concise captions, and are therefore retained in the dataset.

Image Quality Assessment. Since images in FTPD-ML are extracted from published PDF documents, their quality is influenced not only by the original cartographic or illustrative quality in the source articles, but also by factors such as PDF compression, page rendering, figure cropping, and parsing procedures. An automated quality inspection was conducted over all 497,951 figure-text samples. The results indicate that no duplicate or near-duplicate samples were detected, and no corrupted images were identified. Low-resolution images account for 5.09% of the entire dataset.

Given that FTPD-ML is designed to preserve the natural quality distribution of scientific figures in real publication environments, no post-processing operations such as super-resolution, deblurring, or image enhancement were applied. Instead, images are retained in their original extracted state. This design avoids potential artifacts or semantic shifts introduced by image enhancement procedures and allows users to selectively construct subsets based on task-specific quality requirements.

Recent studies in image restoration and deblurring have demonstrated that degradation factors such as blur, noise, and compression artifacts can significantly affect visual representation learning and downstream model performance. For example, MB-TaylorFormer v2³⁵ improves image restoration performance under multiple degradation conditions through a multi-branch Transformer architecture; DBLRNet/DBLRGAN³⁶ investigates video deblurring from a spatiotemporal modeling perspective; and MC-Blur³⁷ introduces a large-scale benchmark with diverse blur factors to facilitate analysis of real-world degradation. Inspired by these studies, we treat image quality variation as an important metadata attribute and explicitly document it as part of the usage considerations for FTPD-ML. Future work may further integrate image restoration, deblurring, or quality-aware training strategies to investigate how different quality levels of samples influence multimodal models in cartography-related tasks.

Semantic Consistency Evaluation. To further assess the semantic consistency of the figure-text pairs in the cartographic domain, a human evaluation protocol was designed. Since FTPD-ML contains both map images and other scientific figures from map-related literature, a two-stage evaluation was conducted. The first stage evaluates general semantic dimensions including semantic relevance, visual grounding, information completeness, factual correctness, and linguistic fluency. The second stage performs a finer-grained evaluation on cartography-specific aspects: map theme recognition, spatial relation expression, and map symbol understanding. Three annotators with backgrounds in cartography or geographic information science independently evaluated randomly sampled test samples using a standardized scoring scheme, with inter-annotator calibration conducted prior to formal evaluation. The results indicated that the dataset exhibits consistent semantic alignment across both general and cartography-specific dimensions, confirming the reliability of the constructed figure-text pairs for multimodal research in this domain. Detailed evaluation results are available in the GitHub repository, as noted in the Data Availability section.

Usage Notes

FTPD-ML is designed to provide a reusable data foundation for multimodal research in cartography and spatial information science. The dataset includes both map images and other scientific figures extracted from related literature, such as remote sensing imagery, statistical charts, flow diagrams, experimental result visualizations, and methodological schematics. Accordingly, the dataset can support a wide range of multimodal applications, including scientific image understanding, image-text retrieval, image captioning, scientific document analysis, and cartography-oriented vision-language research.

For different downstream tasks, users may select appropriate textual fields according to the required level of granularity. Figure-Caption retains the complete semantic information of the original captions and is suitable for image captioning, multimodal semantic modeling, and tasks requiring high information completeness. First-Caption and Single-Caption provide shorter and more focused textual descriptions, making them suitable for image-text matching, retrieval, or training-efficient scenarios. Multiple-Caption preserves additional

contextual sentences within captions and is better suited for scientific image tasks requiring richer semantic expression. Contextual paragraph text provides complementary contextual information for analyses of how figures support experimental findings, spatial phenomena, or methodological descriptions within scientific articles.

Since not all samples in FTPD-ML are strictly map images, when research objectives focus on tasks such as map theme recognition, spatial relation expression, map symbol interpretation, or other cartographic semantic tasks, users are recommended to construct domain-specific subsets using domain labels, image-type labels, or manually curated filtering results. For general multimodal learning across heterogeneous scientific image types, the full dataset can be directly used to compare the characteristics of different image categories and textual granularities across multimodal applications.

The released FTPD-ML dataset consists solely of figure-text pairs extracted from publications licensed under CC-BY, ensuring full compliance with open license terms for redistribution and reuse. Prior to release, all source publications were verified against the Crossref database³⁸, and no retracted works were found at the time of verification. Given that publication statuses may change over time, we advise users to consult authoritative bibliographic sources for the latest status when performing downstream analyses.

Users should also be aware of variations in sample quality. Although this study has conducted technical validation for image extraction, caption alignment, contextual linking, and figure-text semantic association, the dataset may still contain a small number of low-resolution images, scanning artifacts, caption truncations caused by complex layouts, and partial semantic loss in multi-panel figures. For applications requiring high image quality or precise semantic interpretation, additional manual inspection and task-specific sample selection are recommended.

Users should select appropriate subsets according to the objectives of their downstream applications, as different figure types and textual representations provide complementary semantic information. In particular, for map-related research, automatic lexical overlap-based metrics may not adequately capture whether map themes, spatial relations, or symbol semantics are correctly represented. Therefore, for studies focusing on cartographic semantic understanding, it is recommended to complement automatic evaluation with human assessment or domain-specific evaluation protocols.

To demonstrate the practical utility of the constructed dataset, we conducted representative downstream experiments including figure-text matching and image captioning using samples from the figure-text corpus. The results consistently showed improved performance after training on FTPD-ML, confirming its effectiveness as a multimodal resource. Detailed experimental configurations and quantitative results are available in the FTPD-ML GitHub repository.

Overall, FTPD-ML is positioned as a foundational multimodal resource for cartography and spatial information science, supporting dataset reuse, subset selection, multimodal applications, and reproducible research, rather than representing a benchmark dedicated to a single narrowly defined task. Users may flexibly construct task-specific subsets along multiple dimensions, including textual granularity, image type, and Open-license status.

Data availability

The FTPD-ML dataset is publicly available through the Zenodo repository (<https://doi.org/10.5281/zenodo.21754254>). It contains 75,702 figure-text pairs originating from CC-BY licensed map-related publications, including figure images, associated textual descriptions, and bibliographic metadata. To comply with copyright restrictions, records derived from publications without eligible redistribution licenses are represented by metadata only, and their original figures and textual content are not redistributed.

Code availability

The source code used for dataset construction and technical validation is publicly available in the FTPD-ML GitHub repository (<https://github.com/Laven-scnu/FTPD-ML>). It includes the data processing workflow, figure-text extraction scripts, and the technical validation code for image captioning and image matching, facilitating reuse and reproducibility.

Received: 11 December 2025; Accepted: 27 August 2026;

Published: xx xx xxxx

References

1. Niu, Q. *et al.* From text to multimodality: Exploring the evolution and impact of large language models in medical practice. *arXiv preprint arXiv:2410.01812* (2024).
2. Chen, Z. *et al.* Evolution and prospects of foundation models: From large language models to large multimodal models. *Comput. Mater. Contin.* **80**, 2 (2024).
3. Liang, Z. *et al.* A survey of multimodal large language models. In *Proceedings of the 3rd International Conference on Computer, Artificial Intelligence and Control Engineering (CAICE '24)* 405–409 (Association for Computing Machinery, New York, NY, USA, 2024).
4. Ghosh, A., Acharya, A., Saha, S., Jain, V. & Chadha, A. Exploring the frontier of vision-language models: A survey of current methodologies and future directions. *arXiv preprint arXiv:2404.07214* (2024).
5. Dorkenwald, M., Barazani, N., Snoek, C. G. M. & Asano, Y. M. Pin: Positional insert unlocks object localisation abilities in VLMs. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition* 13548 (2024).
6. Gadre, S. Y. *et al.* Datacomp: In search of the next generation of multimodal datasets. In *Advances in Neural Information Processing Systems* **36** (2023).
7. Gandhi, A., Adhvaryu, K., Poria, S., Cambria, E. & Hussain, A. Multimodal sentiment analysis: A systematic review of history, datasets, multimodal fusion methods, applications, challenges and future directions. *Inf. Fusion* **91** (2023).
8. Zhang, Q. *et al.* Multimodal fusion on low-quality data: A comprehensive survey. *arXiv preprint arXiv:2404.18947* (2024).

9. Liu, Y. *et al.* Mminstruct: A high-quality multi-modal instruction tuning dataset with extensive diversity. *Sci. China Inf. Sci.* **67**, 12 (2024).
10. He, M. *et al.* Efficient multimodal learning from data-centric perspective. *arXiv preprint arXiv:2402.11530* (2024).
11. Boukhers, Z. & Bouabdallah, A. Vision and natural language for metadata extraction from scientific PDF documents: A multimodal approach. In *Proceedings of the 22nd ACM/IEEE Joint Conference on Digital Libraries (JCDL '22)* Article 6, 1–5 (Association for Computing Machinery, New York, NY, USA, 2022).
12. Hu, A. *et al.* MPLUG-PaperOwl: Scientific diagram analysis with the multimodal large language model. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)* 6929–6938 (Association for Computing Machinery, New York, NY, USA, 2024).
13. Srinivasan, K., Raman, K., Chen, J., Bendersky, M. & Najork, M. WIT: Wikipedia-based image text dataset for multimodal multilingual machine learning. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '21)* 2443–2449 (Association for Computing Machinery, New York, NY, USA, 2021).
14. Lin, C.-C., Lin, K., Wang, L., Liu, Z. & Li, L. Cross-modal representation learning for zero-shot action recognition. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition* 19978 (2022).
15. Liu, T., Lam, K.-M., Zhao, R. & Qiu, G. Deep cross-modal representation learning and distillation for illumination-invariant pedestrian detection. *IEEE Trans. Circuits Syst. Video Technol.* **32**, 1 (2021).
16. Zhang, J., Huang, J., Jin, S. & Lu, S. Vision-language models for vision tasks: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **46**, 8 (2024).
17. Singh, A. *et al.* FLAVA: A foundational language and vision alignment model. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition* 15638 (2022).
18. Li, Z. *et al.* MMSci: A multimodal multi-discipline dataset for PhD-level scientific comprehension. *arXiv arXiv:2407.04903* (2024).
19. Song, H., Xu, H., Wang, Z., Wang, Y. & Li, J. A multidisciplinary multimodal aligned dataset for academic data processing. *Sci. Data* **12**, 1 (2025).
20. Li, L. *et al.* Multimodal ArXiv: A dataset for improving scientific comprehension of large vision-language models. *arXiv preprint arXiv:2403.00231* (2024).
21. Otović, E. *et al.* Intra-domain and cross-domain transfer learning for time series data—How transferable are the features? *Knowl.-Based Syst.* **239**, 107976. C. (2022).
22. Deng, W., Xu, Q., Li, S., Wang, X. & Huang, Z. Cross-domain automatic modulation classification using multimodal information and transfer learning. *Remote Sens* **15**, 15 (2023).
23. Cheng, D. *et al.* On domain-adaptive post-training for multimodal large language models. *arXiv preprint arXiv:2411.19930* (2024).
24. Zhang, S. *et al.* A multimodal biomedical foundation model trained from fifteen million image-text pairs. *NEJM AI* **2**, 1 (2025).
25. Feng, D., He, X. & Peng, Y. MKVSE: Multimodal knowledge enhanced visual-semantic embedding for image-text retrieval. *ACM Trans. Multimedia Comput. Commun. Appl.* **19**, 5 (2023).
26. Che, N. *et al.* Multimodality-guided visual-caption semantic enhancement. *Comput. Vis. Image Underst.* **249** (2024).
27. Gruber, J. Markdown: Syntax. *Daring Fireball* <https://daringfireball.net/projects/markdown/> (2004).
28. Piwowar, H. *et al.* The state of OA: A large-scale analysis of the prevalence and impact of open access articles. *PeerJ* **6**, e4375 (2018).
29. Adobe Systems Incorporated. *PDF Reference, Sixth Edition: Adobe Portable Document Format Version 1.7* (Adobe Systems Incorporated, 2006).
30. Llama Index. LlamaParse: A GenAI-native solution for document parsing. <https://www.llamaindex.ai/llamaparse> (2025).
31. Mathpix Inc. Mathpix OCR and convert API. <https://mathpix.com/ocr> (2025).
32. Hsu, T.-Y., Giles, C. L. & Huang, T.-H. K. SciCap: Generating captions for scientific figures. *arXiv preprint arXiv:2110.11624* (2021).
33. Liang, M., Li, H., Zhu, Y. & Zhang, J. FTPD-ML [Dataset]. *Zenodo* <https://doi.org/10.5281/zenodo.21754254> (2026).
34. Radford, A. *et al.* Learning transferable visual models from natural language supervision. In *ICML 2021* 8748–8763 (2021).
35. Jin, Z., Qiu, Y., Zhang, K., Li, H. & Luo, W. MB-TaylorFormer V2: Improved multi-branch linear transformer expanded by Taylor formula for image restoration. *IEEE Trans. Pattern Anal. Mach. Intell.* **47**, 5990–6005 <https://doi.org/10.1109/TPAMI.2025.3559891> (2025).
36. Zhang, K., Luo, W., Yu, Y., Ren, W. & Li, H. Adversarial spatio-temporal learning for video deblurring. *IEEE Trans. Image Process.* **28**, 291–301 (2018).
37. Zhang, K. *et al.* MC-Blur: A comprehensive benchmark for image deblurring. *IEEE Trans. Circuits Syst. Video Technol.* **34**, 3755–3767 (2023).
38. Hendricks, G. *et al.* Crossref: A DOI registration agency and scholarly metadata infrastructure. *Crossref* (2020).
39. Forrester, A. B., Moritz, M. A. & Stephens, S. L. Landscape-scale vegetation change following fire in Point Reyes, California, USA. *Fire Ecol* **7**, 114–128 <https://doi.org/10.4996/fireecology.0702114> (2011).
40. Guo, H. *et al.* Study of RADARSAT-2 synthetic aperture radar data for observing sensitive factors of global environmental change. *J. Appl. Remote Sens.* **8**, 084593 <https://doi.org/10.1117/1.JRS.8.084593> (2014).

Acknowledgements

The authors acknowledge the use of DeepSeek for language polishing of the manuscript. Figures 2, 4, and 5 are images created using the Image2 tool according to our instructions. All scientific content, analyses, and interpretations remain the sole responsibility of the authors.

Author contributions

The contributions of the authors are as follows: Ming Liang and Hui Li were responsible for data production and quality validation; Yunqiang Zhu provided overall supervision and guidance; Jinqiu Zhang designed the experiments.

Funding

This study was funded by the National Key Research and Development Program of China (No. 2025YFE0211500), the Chinese Academy of Sciences under Grant XDB0740000, and State Key Laboratory of Resources and Environment Information System under Grant KPI009.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Y.Z. or J.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.